

The Pursuit of a Glioblastoma Cure: A Comprehensive Analysis of Molecular Ontogenies, Microenvironmental Architecture, and Next-Generation Therapeutic Modalities

Author: SAMUELSON G¹

Independent Researcher

Abstract:

Glioblastoma remains the most aggressive and lethal primary malignant brain tumor in adults, with a 5-year survival rate languishing below 10%. Despite maximal safe resection, fractionated radiotherapy, and temozolomide chemotherapy, inevitable recurrence is driven by profound intra-tumoral heterogeneity, the restrictive blood-brain barrier (BBB), and a highly immunosuppressive tumor microenvironment (TME). This comprehensive review synthesizes recent advancements in glioblastoma molecular taxonomy (WHO CNS5), highlighting the prognostic significance of IDH-wildtype status, TERT promoter mutations, and EGFR amplification. We delineate the spatial-epigenetic resistance loop governed by glioblastoma stem cells (GSCs) across distinct microenvironmental niches, alongside emerging metabolic vulnerabilities. Furthermore, we evaluate next-generation therapeutic modalities currently traversing clinical trials, including oncolytic virotherapy, multi-target CAR T-cell therapy, and peptide vaccines. Biophysical disruption techniques, such as Tumor Treating Fields (TTFields) and ultrasound-mediated BBB opening, alongside nanomedicine and sonodynamic therapy, are presented as mechanisms to bypass pharmacokinetic barriers. Finally, the integration of AI-driven radiomics, liquid biopsy monitoring, and the GBM-AGILE adaptive trial framework are discussed as essential paradigms for precision neuro-oncology. Ultimately, a curative approach requires patient-specific, combinatorial strategies that dismantle the TME, exploit metabolic vulnerabilities, and enforce durable

immune surveillance.

Keywords: Glioblastoma; Tumor Microenvironment; Immunotherapy; Nanomedicine; Tumor Treating Fields; Epigenetics; Radiomics; Sonodynamic Therapy; Blood-Brain Barrier.

1. Introduction: Epidemiology, Clinical Imperative, and Conventional Therapeutics

Glioblastoma stands as one of the most formidable adversaries in modern oncology, representing the most common, aggressive, and universally lethal primary malignant brain tumor in the adult population.¹ Accounting for approximately half of all adult gliomas, its incidence is persistently documented at roughly 3.2 per 100,000 individuals annually in the United States alone.² Characterized by rapid progression, profound neurological morbidity, and a near-inevitable propensity for recurrence, glioblastoma has historically resisted the most aggressive and sophisticated therapeutic interventions.¹ Despite decades of intensive, multidisciplinary research and the continuous evolution of neuro-oncological protocols, the prognostic outlook remains bleak. The current standard of care—which includes maximal safe surgical resection, followed by fractionated external beam radiotherapy and concurrent chemotherapy with the alkylating agent temozolomide (TMZ)—yields a median overall survival (OS) of merely 12 to 15 months for newly diagnosed patients.¹ The two-year survival rate languishes below 30%, and the five-year survival milestone is achieved by fewer than 10% of patients.¹ Upon the almost certain event of disease recurrence, median survival plummets to approximately six to ten months, a phase where salvage therapies offer primarily palliative benefits rather than curative potential.¹

The profound recalcitrance of glioblastoma to conventional therapeutics is driven by a confluence of unique anatomical, biological, and microenvironmental obstacles. The central nervous system (CNS) is shielded by the blood-brain barrier (BBB), an intricate network of specialized endothelial cells, tight junctions, and astrocyte foot processes that acts as an evolutionary safeguard.⁷ While essential for protecting the brain from circulating toxins, the BBB serves as a formidable pharmacokinetic shield, severely restricting the penetration of systemic antineoplastic agents and rendering many highly potent chemotherapeutics completely ineffective.⁷ Furthermore, glioblastoma is defined by extraordinary inter- and intra-tumoral heterogeneity.² This means that any single therapeutic modality is likely to eradicate only a specific subpopulation of susceptible

cancer cells, leaving behind resistant clones that rapidly repopulate the tumor mass.⁷

1.1 The Limitations of Conventional Modalities

Surgery remains the hallmark initial therapeutic approach, with the extent of resection (EOR) acting as one of the most critical treatment-related predictors of positive patient outcomes.¹ However, surgical management is inherently limited by the tumor's highly infiltrative nature. Glioblastoma cells invade the surrounding healthy brain parenchyma via hairlike tentacles, making complete microscopic resection an anatomical impossibility without causing catastrophic neurological damage.¹⁰ Tumors located within the eloquent cortex—areas responsible for critical functions such as speech, motor control, and sensory processing—pose an acute surgical challenge.⁹ To maximize the EOR while preserving neurological function, surgeons increasingly rely on advanced fluorescent agents, most notably 5-aminolevulinic acid (5-ALA), to enhance the intraoperative visualization of tumor margins.⁹ In cases of tumor recurrence, reoperation can be considered; evidence suggests a survival benefit in selected patients, particularly those under the age of 60 with a Karnofsky Performance Status (KPS) of 70 or higher, though it carries a reported 8% risk of permanent new neurological deficits.⁹

Radiotherapy, standardly delivered at a median dose of roughly 36 to 62 Gy (typically 2 Gy/day), faces similar limitations.⁹ While it induces vital DNA double-strand breaks in rapidly dividing cells, its efficacy is constrained by the strict tolerance limits of healthy brain tissue, with high-volume re-irradiation carrying a severe risk of radiation necrosis.⁹ Chemotherapeutically, temozolomide and carmustine wafers are standardly employed, but their success is heavily dependent on the tumor's inherent molecular profile, specifically the methylation status of the O6-methylguanine-DNA methyltransferase (MGMT) promoter.¹ The pursuit of a definitive cure requires an absolute departure from these monolithic treatment strategies. The future of glioblastoma therapeutics lies in dynamic, combinatorial approaches that simultaneously address spatial heterogeneity, bypass the physical barrier of the BBB, and subvert the immunosuppressive architecture of the tumor microenvironment.

2. Methodology of Review

A comprehensive review of recent neuro-oncology literature was conducted, focusing heavily on publications from 2018 through 2026. The literature search prioritized studies detailing WHO CNS5 classification updates, novel molecular biomarkers, advanced

adaptive clinical trials (such as the GBM-AGILE platform), and modern imaging diagnostics including the RANO 2.0 criteria.² Furthermore, peer-reviewed articles, real-world clinical trial data, and consensus guidelines provided by the Society for Neuro-Oncology (SNO) and the European Association of Neuro-Oncology (EANO) were systematically synthesized to accurately evaluate both current standards of care and emerging next-generation therapeutic modalities.

3. Molecular Taxonomy and the WHO 2021 CNS5 Classification

The landscape of neuro-oncology experienced a seismic paradigm shift with the publication of the 2021 fifth edition of the World Health Organization (WHO CNS5) Classification of Tumors of the Central Nervous System. This update officially integrated complex molecular and genetic profiles into the foundational taxonomy of gliomas, superseding decades of reliance on purely morphological and histological definitions.² Historically, the diagnosis of glioblastoma required the microscopic observation of pseudopalisading tumor necrosis and/or microvascular proliferation.³ The WHO CNS5 classification radically redefines the disease: the term "glioblastoma" is now strictly and exclusively reserved for adult-type, diffuse astrocytic gliomas that are isocitrate dehydrogenase (IDH)-wildtype.¹³

This segregation rectifies a long-standing clinical ambiguity. Tumors harboring IDH1 or IDH2 mutations, which were previously termed "secondary glioblastomas" when exhibiting high-grade necrotic features, are now reclassified entirely as "Astrocytoma, IDH-mutant, CNS WHO grade 4".⁸ This distinction is critical because IDH-mutant tumors possess distinctly different epigenetic ontogenies and generally confer a more favorable prognostic trajectory compared to the inherently hyper-aggressive IDH-wildtype glioblastomas.¹

Crucially, the WHO CNS5 guidelines mandate that an IDH-wildtype diffuse astrocytoma can be formally diagnosed as a Glioblastoma, CNS WHO grade 4, even in the complete histological absence of classic high-grade features (necrosis or microvascular proliferation), provided that specific, aggressive molecular hallmarks are present.¹³ This acknowledges the clinical reality that tumors harboring these genetic aberrations will rapidly behave with grade 4 aggressiveness, regardless of their initial microscopic appearance.¹³

Molecular Diagnostic Marker	Clinical and Diagnostic Significance in WHO CNS5
IDH-Wildtype Status	The foundational requisite for a glioblastoma diagnosis. Indicates the absence of mutations in IDH1 or IDH2 genes, distinguishing it from lower-grade or IDH-mutant astrocytomas. ²
TERT Promoter Mutation	Involves mutations at hotspots C228T or C250T (c.-124 and c.-146). Results in the maintenance of telomere length, allowing malignant cells to repress cellular senescence and proliferate endlessly. Present in 50-60% of diffuse gliomas. Its presence in an IDH-wildtype astrocytoma is sufficient to assign a grade 4 glioblastoma diagnosis. ²
EGFR Gene Amplification	Epidermal growth factor receptor amplification leads to the severe deregulation of tumor cell proliferation via overactivation of pro-oncogenic signaling pathways (e.g., the EGFRvIII mutation). Frequent co-occurrence with PTEN and TERT mutations often results in highly multifocal tumors. Sufficient for a grade 4 diagnosis. ²
+7 / -10 Chromosome Alteration	The combination of whole chromosome 7 gain (trisomy 7) and whole chromosome 10 loss (monosomy 10) is a frequent copy-number aberration. Its presence is sufficient to upgrade a diffuse astrocytoma to a grade 4

	glioblastoma. ¹³
CDKN2A/B Homozygous Deletion	Loss of function of these crucial tumor suppressor genes promotes uncontrolled cell growth. While heavily utilized to grade IDH-mutant astrocytomas to grade 4, it also indicates profound aggressiveness in the glioblastoma molecular profile. ²
PTEN Loss	Loss of the Phosphatase and Tensin Homolog (PTEN) suppressor gene accelerates uncontrolled proliferation and is frequently observed alongside EGFR amplification, driving the pathogenesis of treatment-resistant phenotypes. ²
MGMT Promoter Methylation	While not a defining WHO diagnostic criterion, it is an essential predictive biomarker. Methylation silences the DNA repair enzyme MGMT, predicting a superior therapeutic response to alkylating agents like temozolomide. Patients with unmethylated MGMT promoters typically suffer worse outcomes and derive minimal benefit from standard chemotherapy. ¹

The codification of these markers reveals a profound underlying clinical insight: the true driver of glioblastoma's lethality is its underlying molecular engine. Identifying these alterations provides immediate therapeutic stratification, guiding oncologists away from futile interventions and toward clinical trials tailored to specific genetic vulnerabilities.

4. The Pathobiology of Resistance: Glioblastoma Stem Cells

At the core of glioblastoma's resistance mechanisms lies a highly plastic and elusive subpopulation of cells known as glioblastoma stem cells (GSCs), or glioblastoma-initiating cells.² Originating potentially from neural stem cells in the subventricular zone, these precancerous cells evolve into heterogeneous tumors by activating multiple gliomagenesis programs.³ GSCs possess the defining characteristics of stemness: profound self-renewal capacity, multipotency, and the unique ability to regenerate vast tumor heterogeneity from a minimal surviving cellular remnant.²

GSCs are fundamentally responsible for the disease's relentless progression and inevitable recurrence. They exhibit distinctly enhanced DNA repair mechanisms and profound metabolic adaptations that shield them from the DNA-damaging effects of fractionated radiotherapy and the alkylating stress of temozolomide.¹⁰ Furthermore, GSCs do not exist in isolation; they actively engineer and reside within specialized microenvironmental niches—particularly perivascular and hypoxic zones—that physically and biochemically sustain their stemness.¹⁰ Recent molecular discoveries highlight that signaling pathways such as Notch, Wnt/ β -catenin, Hedgehog, and STAT3-PARN are critical regulators of GSC maintenance.¹⁰ Additionally, factors like TFPI2 and HML-2 facilitate both angiogenesis and immune evasion, essentially wrapping the GSCs in an impenetrable biological armor.¹⁰ Eradicating the bulk tumor mass while leaving the GSC subpopulation intact is the primary reason why standard therapies ultimately fail; the tumor simply regrows, often with acquired resistance derived from the survival of these progenitor cells.⁷

5. The Spatial-Epigenetic Resistance Loop and Longitudinal Evolution

Glioblastoma's resistance to radiotherapy and chemotherapy is not merely genetic; it is orchestrated by a highly dynamic interplay between the spatial architecture of the tumor and its epigenetic landscape.¹⁹ This phenomenon, defined as the spatial-epigenetic resistance loop, delineates how distinct microenvironmental niches impose unique evolutionary pressures that reprogram the epigenetic state of resident cells.¹⁹ The epigenome—encompassing DNA methylation, histone modification, chromatin remodeling, and the transfer of non-coding RNAs via exosomes—provides the rapid adaptability required for tumor survival under acute therapeutic stress without altering the underlying genetic sequence.⁷

Research indicates that the glioblastoma ecosystem is geographically partitioned into distinct functional niches, each leveraging specific epigenetic mechanisms to evade targeted destruction:

1. **The Hypoxic Core Niche:** Deep within the poorly vascularized core of the tumor, profound hypoxia activates the HIF-SIRT (Hypoxia-Inducible Factor and Sirtuin) axis.¹⁹ The severe lack of oxygen intrinsically inhibits TET DNA demethylases, leading to the widespread hypermethylation and subsequent silencing of pro-apoptotic genes.¹⁹ This epigenetic configuration traps the resident GSCs in a highly resistant, quiescent state. Because standard radiotherapy relies heavily on the presence of oxygen to generate DNA-damaging reactive oxygen species (free radicals), cells in the hypoxic core are exceptionally radioresistant.¹⁹ The therapeutic implication is profound: utilizing Histone Deacetylase (HDAC) inhibitors specifically within this niche is hypothesized to artificially impair the DNA repair mechanisms these cells rely upon, thereby re-sensitizing them to radiation.¹⁹
2. **The Invasive Edge Niche:** At the tumor's periphery, where malignant cells actively infiltrate the healthy brain parenchyma, a different epigenetic program is activated. Cells in this niche utilize EZH2-mediated histone methylation, specifically the H3K27me3 mark.¹⁹ The overexpression of EZH2 stabilizes a chromatin state that drives the proneural-mesenchymal transition (PMT).¹⁹ This phenotypic shift endows the cells with profound motility, enhanced invasiveness, and an increased capacity for radiation evasion.¹⁹ Consequently, maximal surgical resection frequently fails because these mesenchymal-shifted cells have already epigenetically reprogrammed themselves to migrate far beyond the visible radiological margins.¹¹ Deploying EZH2 inhibitors specifically targeting the invasive edge could suppress this mesenchymal shift, potentially confining the tumor for more effective local therapy.¹⁹
3. **The Perivascular Niche (PVN):** Adjacent to the chaotic, hyper-proliferative tumor vasculature, GSCs reside in a specialized microenvironment that continuously sustains their stemness.¹⁰ This niche utilizes HDAC-mediated DNA repair pathways alongside BRD4-driven super-enhancer mechanisms.¹⁹ These super-enhancers, marked by extensive H3K27ac modifications, drive the relentless expression of critical oncogenes such as SOX2 (essential for stemness) and EGFR.¹⁹ To dismantle these perivascular stemness programs, Bromodomain and Extraterminal (BET) inhibitors are currently under investigation to directly target BRD4, thereby collapsing the three-dimensional genomic architecture that the GSCs rely upon for

survival.¹⁹

5.1 Longitudinal Epigenetic Evolution Under Treatment

When an IDH-wildtype glioblastoma is subjected to the alkylating stress of temozolomide and concurrent radiation, the tumor undergoes significant longitudinal genomic and epigenomic evolution.²¹ In contrast to IDH-mutant astrocytomas, which generally show a consistent global decrease in methylation following standard treatment, IDH-wildtype glioblastomas display highly heterogeneous, patient-specific epigenetic shifts.²¹ This plasticity allows glioblastomas to undergo epigenetic class switching or even morph into a sarcomatous transformation.²¹

Critically, the DNA methylation profile of the initial, treatment-naïve tumor can predict its evolutionary trajectory. High levels of methylation at multiple CpG sites in the upstream promoter regions of the MGMT and KCNQ1DN genes dictate the likelihood that the tumor will develop somatic hypermutation upon recurrence following temozolomide treatment.²¹ Furthermore, a survival-associated DNA methylation evolution signature involving 347 specific CpG sites has been identified.²¹ Patients whose recurrent tumors shift toward global hypermethylation at these sites (Group A) experience surprisingly favorable clinical outcomes, with a median survival of 34.9 months.²¹ Conversely, patients whose tumors undergo a hypomethylation shift (Group C) face a dismal median survival of only 16.3 months, a trajectory often enriched by the acquisition of TP53 and RB1 mutations.²¹ Understanding these trajectories dictates that static, universal treatments are biologically obsolete against an actively evolving epigenome; continuous profiling through liquid biopsies and spatial multi-omics is essential to deploy adaptive, combinatory regimens.¹⁹

6. Metabolic Vulnerabilities, Glycan Shields, and RNA Inhibition

Beyond the epigenome, glioblastoma exhibits profound metabolic reprogramming. Recent *in silico* and clinical investigations have highlighted specific nutrient dependencies, notably the upregulation of glutamine transporters such as SLC25A13 and SLC38A2, which correlate significantly with reduced overall survival. Expanding on metabolic exploitation, clinical trials are actively evaluating gallium maltolate (GaM), an oral agent that mimics iron; glioblastoma cells lethally absorb gallium, which fatally disrupts their iron-dependent growth pathways. Furthermore, non-coding RNAs that drive tumor progression are becoming actionable targets. The small-molecule inhibitor

linifanib is being utilized to successfully suppress miR-10b, a critical RNA promoter of glioblastoma cell survival. Additionally, the experimental medication MT-125 is demonstrating the ability to target cellular molecular "motors", rendering previously resistant glioblastoma subtypes sensitive to standard chemoradiation.

7. The Immunosuppressive Tumor Microenvironment and T-Cell Exhaustion

The profound failure of conventional immunotherapies, particularly systemic immune checkpoint inhibitors, in the treatment of glioblastoma is fundamentally rooted in the tumor's unparalleled ability to engineer a hostile, immunosuppressive tumor microenvironment (TME).²² Glioblastoma is widely and accurately characterized as an immunologically "cold" tumor.²³ This coldness is defined by a severe paucity of tumor-infiltrating effector T lymphocytes and an overwhelming presence of protumorigenic, suppressive immune cells.²³

The TME in glioblastoma is not merely a bystander but a highly organized, suppressive network dominated by Tumor-Associated Macrophages (TAMs), Myeloid-Derived Suppressor Cells (MDSCs), and regulatory T cells (Tregs).⁸ The central nervous system is partially immune-privileged, and the blood-brain barrier severely restricts the initial infiltration of systemic lymphocytes and antigen-presenting cells (APCs).²³ However, the microglia and peripheral macrophages that do successfully infiltrate the tumor bed are rapidly hijacked. Based on TME signaling, these macrophages polarize from a tumor-inhibiting M1 phenotype into a tumor-promoting M2 phenotype.²⁷ These M2-polarized TAMs, along with recruited Tregs, actively secrete a barrage of highly immunosuppressive cytokines, most notably Interleukin-10 (IL-10) and Transforming Growth Factor-beta (TGF- β).²⁴ These cytokines systematically dismantle the host's anti-tumor response by directly inhibiting the proliferation, recruitment, and cytotoxic functions of CD8+ T cells.²⁴ This dynamic is further exacerbated by the Siglec–sialic acid immune axis, an immunoregulatory pathway that glioblastoma exploits to temper immune responses. Mapping how macrophages co-evolve with glioblastoma cells provides new therapeutic targets to strip the tumor of this protective glycan shield.

Simultaneously, the malignant glioblastoma cells deploy direct, tumor-intrinsic immune evasion strategies.²⁸ One of the most prominent and damaging mechanisms is the robust upregulation and overexpression of Programmed Death-Ligand 1 (PD-L1) on the tumor cell surface.²⁴ Under normal immunological function, CD8+ T cells recognize tumor

antigens via T-cell receptors (TCRs), releasing cytotoxic granules containing perforin and granzyme to induce apoptosis in the cancer cells, while secreting Interferon-gamma (IFN- γ) and Tumor Necrosis Factor-alpha (TNF- α) to inhibit further tumor growth.²⁴ However, when the overexpressed PD-L1 on the glioblastoma cell binds to the PD-1 receptors on the surface of infiltrating T lymphocytes, it triggers an inhibitory signaling cascade that induces severe T-cell exhaustion and functional impairment.²⁴ This exhaustion paralyzes the T cell, preventing the release of perforin and granzymes and shifting the cells into a dysfunctional, pro-tumor state.²⁴

This immune evasion is further compounded by extreme antigenic heterogeneity, the downregulation of Major Histocompatibility Complex (MHC) molecules, and the upregulation of Indoleamine 2,3-dioxygenase (IDO), which depletes local tryptophan essential for T-cell survival.²⁸ Additionally, the extracellular matrix (ECM) of the tumor, rich in fibronectin, tenascin-C, and hyaluronic acid, creates a dense physical barrier that impedes lymphocyte motility.⁸ Consequently, the systemic administration of PD-1/PD-L1 checkpoint inhibitors (such as pembrolizumab or nivolumab) frequently fails in glioblastoma. These drugs rely on releasing the "brakes" on a pre-existing population of exhausted T cells; however, in glioblastoma, the baseline population of effector cells within the CNS is practically nonexistent, rendering the checkpoint blockade therapeutically impotent.²³

8. Next-Generation Immunotherapeutics: Viruses, CAR-T, and Vaccinology

To circumvent the inherent immunoresistance of glioblastoma, oncological research has decisively pivoted toward therapies that artificially force immune infiltration, prime the TME, or introduce synthetically modified, hyper-potent effector cells directly into the tumor architecture.

8.1 Oncolytic Virotherapy (OVT)

Oncolytic virotherapy represents a sophisticated, dual-action paradigm designed to dismantle the TME from within. A highly promising phase 1 clinical trial, detailed in a 2026 publication in *Cell* and led by researchers at Dana-Farber and Mass General Brigham, utilized a genetically modified herpes simplex virus (HSV).²⁵ This oncolytic virus was meticulously engineered to selectively infect and replicate exclusively within the aberrant environment of glioblastoma cells, leaving normal, healthy neural tissue

unharmful.²⁵

The primary mechanism of action involves direct viral oncolysis: the virus enters the malignant cell, hijacks its machinery to replicate massively, and causes physical cellular lysis, thereby releasing thousands of viral progeny to infect adjacent tumor cells.²⁵ However, the second-order, immunological effect of OVT is far more consequential for long-term survival.²² The explosive, necrotic death of virus-infected tumor cells releases a massive payload of previously hidden tumor-associated antigens, alongside highly immunogenic viral pathogen-associated molecular patterns (PAMPs), directly into the TME.²² This highly immunogenic cell death effectively shatters the immunosuppressive shield, forcibly recruiting peripheral immune cells and converting the tumor from immunologically "cold" to "hot".²⁵ Similarly, researchers at MD Anderson are utilizing the Delta-24-RGD engineered oncolytic virus. Crucially, they have identified blood-based biomarkers capable of distinguishing "elite responders"—patients who will derive the most significant survival advantage—early in the treatment cycle, allowing for personalized, additional dosing.

8.2 Chimeric Antigen Receptor (CAR) T-Cell Therapy

CAR T-cell therapy involves extracting a patient's autologous T lymphocytes and genetically engineering them to express chimeric receptors that explicitly target specific surface antigens present on glioblastoma cells.⁶ While systemically delivered CAR T cells struggle immensely to cross the BBB, intracranial delivery—infusing the engineered cells directly into the tumor cavity or the cerebrospinal fluid—has demonstrated remarkable feasibility, safety, and potent local bioactivity.³

Early pioneering trials, such as NCT02208362, evaluated CAR T cells engineered to target the IL13R α 2 receptor. Administered to 65 patients, this approach yielded a 50% disease-control rate and a 23% one-year survival rate, representing a significant milestone for recurrent disease where standard survival is measured in months.³ However, the extreme antigenic heterogeneity of glioblastoma presents a fatal flaw for single-antigen therapies: antigen escape. The tumor rapidly downregulates the targeted protein, allowing the remaining, antigen-negative glioblastoma cells to proliferate unhindered.²⁹

To counteract this, next-generation multi-target CAR T cells have been developed. The INCIPIENT phase 1 trial evaluated CARv3-TEAM-E T cells, designed to simultaneously target both the mutated EGFRvIII and the wild-type EGFR.³⁰ The clinical outcomes were

characterized by dramatic, rapid tumor regressions occurring within mere days of a single infusion, with one patient achieving near-complete radiological tumor regression.³⁰ Similarly, a dual-target CAR T therapy trial at the University of Pennsylvania (NCT05168423) resulted in significant tumor shrinkage in nearly two-thirds of patients, with several surviving well beyond 12 months post-treatment.³ Presented at the 2025 American Society of Clinical Oncology (ASCO) Annual Meeting, updated findings confirmed that nearly two-thirds of patients experienced tumor shrinkage, with multiple patients surviving 12 months or longer post-treatment.⁶

Despite these spectacular initial responses, longitudinal observation via serial biopsies and cerebrospinal fluid sampling revealed a sobering reality: the radiological responses were largely transient, with tumors eventually progressing.³ This transiency highlights a critical pathophysiological insight. While the CAR T cells possess overwhelming initial cytolytic capacity, upon entering the glioblastoma TME, they rapidly fall victim to the harsh, hypoxic, and immunosuppressive conditions (driven by TGF- β and PD-L1), resulting in terminal CAR T-cell exhaustion.²⁹ Consequently, future iterations are focusing on engineering "armored" CAR T cells that co-express supportive cytokines or are genetically edited to resist PD-L1 suppression, ensuring durable intratumoral persistence.²⁹

8.3 Vaccinology and Innate Immune Activation

Active immunotherapies, such as therapeutic cancer vaccines, seek to continuously educate the host's endogenous immune system to recognize and eradicate glioblastoma cells, offering a pathway to durable, long-term immunological memory.³¹

SurVaxM is a highly anticipated peptide vaccine targeting survivin, an anti-apoptotic protein that is highly overexpressed in glioblastoma cells and actively promotes their survival and profound resistance to standard chemotherapy.³² The initial single-arm phase 2a study of SurVaxM in 63 patients with newly diagnosed glioblastoma showed exceptionally promising results.³³ An impressive 93.7% of patients survived one year post-diagnosis (compared to an expected historical baseline of 65%), achieving a median OS of 25.9 months, significantly outperforming standard-of-care metrics.³³ Following these compelling results, the randomized, placebo-controlled phase 2b SURVIVE trial (NCT05163080) completed accrual, combining SurVaxM with standard temozolomide chemotherapy at 11 U.S. sites.³² Interim analyses in 2024 met all prespecified standards for safety and efficacy, allowing the trial to continue seamlessly, heavily supported by the

vaccine's excellent safety profile and maintenance of patient quality of life.³²

Simultaneously, **VBI-1901** is an immunotherapeutic vaccine targeting cytomegalovirus (CMV) antigens, which are anomalously expressed in a high percentage of glioblastoma tumors but remain absent in healthy brain tissue.³⁵ In an ongoing phase 2b randomized controlled study (data presented at ASCO 2024) for recurrent glioblastoma, VBI-1901 demonstrated a 43% disease control rate (3 out of 7 patients) compared to a 0% control rate in the standard-of-care arm.³⁵ Notably, rapid partial responses—including a 67% reduction in tumor size—were observed after just two doses at week six.³⁵ This rapid generation of a disease-controlling immune response prompted the FDA to grant VBI-1901 both Fast Track and Orphan Drug Designations.³⁵

Beyond T-cell-centric approaches, manipulating the innate immune system via Natural Killer (NK) cells offers a mechanism that completely bypasses the need for specific antigen presentation and MHC-class matching.²⁴ A groundbreaking 2025 pilot study (NCT06061809) evaluated a chemotherapy-free regimen combining **ANKTIVA** (nogapendekin alfa inbakicept-pmln, an IL-15 superagonist), PD-L1 t-haNK (a targeted NK cell therapy), and Optune Tumor Treating Fields in patients with highly refractory, recurrent glioblastoma.³⁶ The results were extraordinary: 100% disease control was achieved in the initial cohort of five patients.³⁶ This included two near-complete responses and an increase in absolute lymphocyte counts (ALC) across all patients, successfully reversing the severe lymphopenia caused by prior standard-of-care radiation and chemotherapy.³⁶ This trial specifically targeted second-line glioblastoma patients who had failed the current standard of care, demonstrating the profound ability of the IL-15 agonist to reverse radiation- and chemotherapy-induced lymphopenia.³⁶ The IL-15 agonist essentially supercharges the NK cells, allowing them to identify and destroy tumor cells independent of the suppressive TME, providing a powerful synergistic effect when combined with biophysical tumor disruption.³⁶

9. Biophysical Modalities: Tumor Treating Fields, Focused Ultrasound, and Proton Therapy

The integration of advanced medical physics into neuro-oncology has yielded therapeutic modalities that rely on the biophysical and structural properties of the tumor, effectively bypassing the biochemical and genetic resistance mechanisms that neutralize conventional systemic drugs.

9.1 Tumor Treating Fields (TTFields)

TTFields therapy (marketed commercially as Optune Gio®) utilizes a portable, non-invasive device to deliver alternating, intermediate-frequency (200 kHz) electric fields directly to the brain via arrays placed tightly on the patient's shaven scalp.³⁷ These finely tuned electrical fields exert precise, continuous physical forces that disrupt highly polar cellular components.³⁷ Specifically, TTFields interfere with the polymerization of tubulin, preventing the formation of the mitotic spindle apparatus during cell division.³⁷ This antimitotic effect leads to severe chromosomal instability, disrupted mitosis, and subsequent apoptosis exclusively in rapidly dividing glioblastoma cells, while completely sparing non-dividing healthy neurons.³⁷ Furthermore, TTFields have been proven to induce immunogenic cell death, impair cellular motility and migration, inhibit DNA damage responses, and significantly increase cellular and tissue permeability.³⁷

The integration of TTFields into the global standard of care is supported by robust, large-scale clinical data. The landmark EF-14 phase 3 trial established TTFields as a Category 1 option in the NCCN Guidelines for newly diagnosed glioblastoma.⁴⁰ Recent real-world data from the TIGER study—a massive prospective, non-interventional study enrolling 429 newly diagnosed patients in Germany—corroborated these pivotal findings.³⁹ The TIGER study reported a median OS of 19.6 months and a median progression-free survival (PFS) of 10.2 months, with impressive long-term survival tails of 31.5% at three years and 27.7% at four years.³⁹ This represents the largest prospective, non-interventional study of TTFields therapy in routine clinical care completed to date.³⁹ Crucially, TTFields monotherapy (evaluated in the EF-19 registry) and combination therapy showed no increase in systemic toxicity, underscoring its immense utility as an adjunctive, continuous therapy that maintains patient quality of life.³⁹ Current phase 3 trials, most notably the EF-41/KEYNOTE D58 study (NCT06556563), are evaluating the highly anticipated triple combination of Optune, temozolomide, and pembrolizumab (a PD-1 inhibitor), operating on the hypothesis that the immunogenic cell death and increased permeability caused by TTFields will finally sensitize the glioblastoma to immune checkpoint blockade.³⁸

9.2 Focused Ultrasound (FUS) and Sonodynamic Therapy

As previously established, a major pharmacological bottleneck in glioblastoma treatment is the blood-brain barrier. Focused Ultrasound (FUS) provides an innovative, non-invasive or minimally invasive biophysical solution to temporarily and reversibly open the BBB.⁴¹ The procedure typically involves the intravenous administration of

lipid-encased microbubbles (such as Definity® or perflutren lipid microspheres), followed by the application of low-intensity focused ultrasound targeted with pinpoint MRI precision at the tumor coordinates.⁴² The acoustic pressure waves cause the circulating microbubbles to oscillate and cavitate within the cerebral capillaries.⁴³ This mechanical action stretches and safely disrupts the tight junctions between the endothelial cells, dramatically increasing permeability and allowing large-molecule therapeutics—such as monoclonal antibodies, immune-modulating drugs, CAR T cells, or high-dose chemotherapeutics—to flood directly into the tumor parenchyma.⁴²

Clinical trials have rapidly advanced the safety and efficacy profiles of this technology. Early-stage trials utilizing the NaviFUS system in Taiwan demonstrated that FUS-mediated BBB opening combined with albumin-bound paclitaxel or standard chemotherapy is safe, feasible, and provides a possible survival benefit in patients with recurrent glioblastoma.⁴¹ A 34-patient trial utilizing Insightec's Exabate Neuro system confirmed the safety of repeated, monthly BBB opening prior to temozolomide cycles.⁴³ In the United States, the SONOBIRD phase 3 study is evaluating the SonoCloud-9, a novel surgically implanted device equipped with nine ultrasound emitters.⁴² Implanted during resection for recurrent glioblastoma, the SonoCloud-9 allows for regular, on-demand activation to repeatedly disrupt the BBB prior to the administration of systemic therapies.⁴²

Beyond simple barrier opening, focused ultrasound is driving the emerging field of Sonodynamic Therapy (SDT). By administering oral sonosensitizers like 5-ALA followed by low-intensity ultrasound, SDT creates highly localized reactive oxygen species (ROS) that induce targeted tumor cell death. Early clinical trials for recurrent high-grade gliomas utilizing SDT have extended median overall survival to a highly promising 15.7 months.⁴⁶ In a recent phase 1/2 clinical trial, SDT safely extended the median overall survival to 15.7 months in recurrent high-grade gliomas, representing a greater than twofold increase over historical expectations. Furthermore, for elderly populations (over 65) who typically tolerate standard radiation poorly, recent Mayo Clinic trials leveraging short-course hypofractionated proton beam therapy have extended median survival to 13.1 months, nearly doubling historical expectations.

10. Nanomedicine and Targeted Drug Delivery Systems

In tandem with macroscopic biophysical disruption, nanoscale engineering offers unprecedented molecular precision in drug delivery. Nanotherapy aims to encapsulate

highly toxic chemotherapeutics within sophisticated delivery vectors that can stealthily navigate the systemic circulation, avoid premature immune clearance, cross the BBB via specialized cellular transport mechanisms, and release their payload strictly within the acidic or hypoxic tumor microenvironment.⁴⁷

Transporting agents across the intact BBB requires exploiting specific endothelial mechanisms. Nanocarriers are engineered to utilize receptor-mediated transcytosis (binding to specific endothelial receptors), carrier-mediated transcytosis, or adsorption-mediated transcytosis (leveraging electrostatic interactions between positively charged nanoparticles and negatively charged cell membranes) to actively ferry drugs into the brain while avoiding rapid efflux pumps that typically eject small molecules back into the bloodstream.⁴⁷

Nanotherapeutic Platform	Mechanism of Action and Delivery Status
Silica-Based Nanoparticles	Ultra-small large-pore silica nanoparticles and mesoporous silica nanoparticles (MSNs) are heavily engineered to carry temozolomide and chloroquine directly to glioblastoma cells. Their porous structure protects the payload from systemic degradation, significantly overcoming localized cellular efflux and resistance mechanisms. ⁴⁷
Exosomes and Extracellular Vesicles	Biomimetic delivery vehicles utilizing naturally occurring blood-based or embryonic stem cell-derived exosomes. Engineered variations target specific mutations (e.g., anti-EGFRvIII targeted exosomes) to induce apoptosis. Chimeric exosomes can be functionalized with STING activation for personalized

	immunotherapy or disguised with macrophage membranes to enhance targeted sonodynamic therapy. ⁴⁷
Liposomal and Polymer Formulations	Lipid-based nanoformulations, such as pegylated liposomal doxorubicin, utilize a polyethylene glycol (PEG) coating to extend circulatory half-life and shield the payload. Biomimetic liposomes infused with macrophage cell membranes are actively undergoing phase 2 evaluation for targeted delivery following radiotherapy. ⁴⁷
Iron Oxide Nanoparticles (NanoTherm™)	Currently the first and only clinically approved nanoparticle-based therapy for glioblastoma (approved in Europe). Iron oxide nanoparticles are injected directly into the tumor bulk. The application of an external alternating magnetic field generates intense localized hyperthermia, thermally ablating the tumor tissue and acting as a profound radioenhancer for subsequent radiotherapy. ⁴⁷

While these platforms solve the issue of systemic toxicity and targeted delivery, scaling up production and high costs remain significant implementation hurdles.⁴⁷ To further enhance delivery, researchers are coupling nanoparticles with Convection-Enhanced Delivery (CED).⁴⁷ In CED, catheters are stereotactically placed directly into the brain, and nanocarriers are infused under continuous positive pressure. This technique physically bypasses the BBB and overcomes traditional diffusion limits, bathing the infiltrative, migratory edges of the tumor in high-concentration cytotoxic agents without exposing peripheral organs to systemic toxicity.⁴⁷

11. Adaptive Clinical Trial Frameworks: The GBM-AGILE Paradigm

The historical inefficiency of sequential, rigidly structured, slow-moving clinical trials has been a major systemic impediment to glioblastoma drug discovery. The establishment of the GBM-AGILE (Glioblastoma Adaptive, Global, Innovative Learning Environment) trial fundamentally restructured this developmental paradigm.⁴⁸ Sponsored by the Global Coalition for Adaptive Research (GCAR), GBM-AGILE (NCT03970447) is an international, seamless phase 2/3 Bayesian response-adaptive randomization platform registration trial.⁴⁸

Operating under a single Master Protocol, GBM-AGILE evaluates multiple investigational drugs concurrently against a common internal control arm (temozolomide for newly diagnosed patients; lomustine for recurrent disease).⁴⁹ The trial matches these therapies to specific prospectively identified patient subtypes (e.g., newly diagnosed methylated, newly diagnosed unmethylated, and recurrent disease).⁵⁰ The adaptive Bayesian statistical design continuously analyzes patient performance and overall survival (OS) data in real-time.⁵⁰ If an investigational arm demonstrates a high Bayesian predictive power of superiority (a probability of benefit $\geq 98\%$ with a Hazard Ratio < 1.00), it rapidly graduates to a Stage 2 fixed randomization expansion cohort suitable for regulatory filing.⁵⁰ Conversely, if the predictive power falls below 25% across all signatures during monthly analyses, the arm is swiftly and definitively terminated for futility.⁵⁰

This brutal but highly efficient mathematical culling mechanism recently provided critical, definitive real-world data on several highly anticipated agents:

- **Regorafenib:** An oral multikinase inhibitor previously thought to be highly promising based on earlier phase 2 data (the REGOMA trial). Within the rigorous structure of GBM-AGILE, regorafenib failed to improve overall survival across all tested signatures.⁵⁰ For NDU patients, the Bayesian-modeled median OS was 14.28 months for regorafenib versus 14.56 months for the control (Hazard Ratio 1.05).⁵⁰ For recurrent disease, the OS was 9.20 months versus 9.69 months for the control (HR 1.07).⁵⁰ Furthermore, regorafenib was associated with significantly increased Grade 3 toxicity.⁵⁰ Consequently, the trial accrued 172 patients before stopping early for limited efficacy, leading to the definitive removal of regorafenib from NCCN guidelines as a treatment option for recurrent glioblastoma.⁵⁰
- **VAL-083 (Dianhydrogalactitol):** A novel, CNS-penetrating DNA-targeting agent uniquely designed to create inter-strand crosslinks at the N7 position of guanine

residues.⁵¹ This unique cytotoxic mechanism was explicitly designed to induce DNA double-strand breaks independent of MGMT promoter methylation, theoretically bypassing standard chemoresistance.⁵¹ Despite robust in vitro and in vivo preclinical rationale, VAL-083 failed to outperform the standard of care in its Stage 1 evaluation within GBM-AGILE.⁵⁴ The Bayesian predictive power fell below the required 0.8 threshold, prompting the sponsor (Kintara Therapeutics) to suspend all development activities and costs related to the agent in this indication.⁵¹

- **Paxalisib and BA-101:** Conversely, the adaptive structure successfully highlighted the necessity for precise molecular matching. Following evaluation in GBM-AGILE, secondary OS data demonstrated a distinct 3.8-month survival improvement for paxalisib specifically in patients with newly diagnosed glioblastoma exhibiting an unmethylated MGMT promoter status.⁴⁶ This targeted, biomarker-driven success led to alignment with the FDA for standard approval pathways for this specific sub-population.⁴⁶ Similarly, BA-101, an investigational small molecule targeting dysregulated nitric oxide signaling, received FDA Orphan Drug Designation after showing preclinical promise in enhancing temozolomide efficacy, representing the next wave of highly targeted inhibitors.⁴⁶

These rigorous outcomes underscore a critical insight: broad-spectrum monotherapies applied universally to unselected, heterogeneous glioblastoma cohorts are statistically and biologically destined to fail. Success is contingent entirely upon molecular stratification and the rapid abandonment of futile mechanisms.

12. AI-Driven Radiomics and Liquid Biopsy Monitoring

The accurate, non-invasive assessment of glioblastoma evolution is undergoing a revolution driven by Artificial Intelligence (AI) and liquid biopsies. AI-driven radiomics involves extracting high-dimensional features from standard MRI and PET scans. Deep learning models (such as convolutional neural networks) can now automatically segment tumors with Dice Similarity Coefficients (DSCs) exceeding 0.90, dramatically reducing radiotherapy planning time. More profoundly, these algorithms can predict molecular biomarkers purely from imaging; studies report accuracies up to 0.96 for predicting IDH status and robust capabilities for predicting MGMT and ATRX mutations.

Concurrently, the neuro-oncology field is advancing the use of liquid biopsies—analyzing cerebrospinal fluid or blood for circulating tumor DNA (ctDNA) and extracellular

vesicles (EVs). These minimally invasive biomarkers provide real-time, longitudinal tracking of the tumor's genetic and epigenetic landscape, offering a vital adjunct to diagnostic protocols by definitively distinguishing true tumor progression from treatment-induced pseudoprogression.

13. Radiographic Response Assessment: The Implementation of RANO 2.0

As novel therapies—particularly immunotherapies, viral oncolytics, and intense biophysical modalities—become integrated into standard clinical practice, the radiographic assessment of tumor response has required dramatic recalibration. The introduction of the Response Assessment in Neuro-Oncology criteria version 2.0 (RANO 2.0) in 2023 directly addresses one of the most confounding challenges in neuro-oncological imaging: pseudoprogression (PsP).⁵³

Pseudoprogression occurs when a post-treatment MRI scan reveals significantly increased contrast enhancement, worsening T2/FLAIR signal, or apparent tumor expansion.⁵³ Under traditional criteria, this unequivocally indicates progressive disease (PD) and treatment failure.⁵³ However, in the context of effective immunotherapy, viral oncolysis, or heavy radiation, this intense enhancement frequently represents a robust inflammatory response, radiation-induced necrosis, increased vascular permeability, and the massive influx of immune cells actively attacking the tumor.⁵³ It is a sign of profound treatment effect, not neoplastic growth. Misinterpreting PsP as true progressive disease historically led to the tragic, premature discontinuation of highly effective, potentially curative therapies.⁵⁶

To systemically mitigate this diagnostic error across all clinical trials and practices, RANO 2.0 implemented sweeping framework changes.⁵⁵

First, it mandates the adoption of the post-radiation scan as the new baseline for future comparisons, acknowledging that immediate post-operative scans do not account for radiation-induced inflammation.⁵⁵ Second, and most critically, it institutes the rigorous rule of "preliminary progressive disease" (preliminary PD).⁵⁵ If a scan shows an increase of $\geq 25\%$ in the sum of products of perpendicular diameters (SPPD) of the target lesions, or the appearance of new enhancing lesions within the first 12 weeks of initial treatment, it cannot be immediately declared as progressive disease if the patient remains clinically stable.⁵⁷ Instead, a mandatory confirmation scan must be obtained at least 4 to 8 weeks later, while the patient carefully continues their current therapy.⁵⁶ Only if this

subsequent confirmation scan demonstrates continued, unabated radiological growth is the patient definitively diagnosed with confirmed PD (at which point the date of progression is backdated).⁵⁶ If the enhancement stabilizes or recedes, the phenomenon is confirmed as pseudoprogression, and the therapy is correctly maintained.⁵⁶

To further refine this distinction non-invasively, advanced multiparametric and volumetric analyses are being actively validated.⁵⁸ Artificial intelligence is being utilized to assess 3D volumetric tumor segmentations, specifically evaluating the complex enhancing-to-T2/FLAIR volume ratios. These AI-guided volumetric assessments act as predictive imaging biomarkers, offering significantly improved sensitivity in distinguishing between the profound, transient edema of pseudoprogression and the dense, cellular proliferation of true progressive disease.⁵⁸

14. Conclusion: Strategic Directives Toward a Curative Paradigm

The pursuit of a cure for glioblastoma has definitively transitioned away from the archaic search for a singular, universally applicable "magic bullet." The sheer complexity of the disease dictates that future success lies entirely in the engineering of sequential, multimodal interventions that acknowledge the tumor not as a static mass of rogue cells, but as a highly adaptive, spatially organized, epigenetically fluid, and immunologically hostile ecosystem. The extensive clinical and biological data analyzed herein establishes several unequivocal, strategic directives for the future of neuro-oncology.

First, standard systemic pharmacological delivery that relies on passive diffusion across an impermeable barrier is fundamentally flawed. The future of chemotherapeutic and immunotherapeutic delivery necessitates active, localized biophysical disruption. Utilizing Focused Ultrasound for precise, reversible BBB fenestration or employing Sonodynamic therapy directly into the tumor, ensures that therapeutic payloads reach the invasive edges of the tumor where recurrence universally originates.

Second, the molecular taxonomy defined by the WHO CNS5 classification must strictly dictate the immediate therapeutic vector. IDH-wildtype tumors driven by TERT promoter mutations and EGFR amplifications require relentless, precision-targeted therapies. Crucially, addressing the spatial-epigenetic resistance loop and metabolic vulnerabilities is paramount. Standard fractionated radiotherapy must be co-administered with niche-specific epigenetic modulators and metabolic disrupters.

Third, the profound immunosuppression of the glioblastoma TME cannot be solved by

the systemic administration of checkpoint inhibitors alone. The microenvironment must be forcibly primed. Modalities like Oncolytic Virotherapy and Tumor Treating Fields are essential not merely for their direct cytocidal effects, but for their unique capacity to induce highly immunogenic cell death. This process shatters the suppressive TME, releasing hidden tumor antigens and converting the tumor from immunologically cold to hot. Once the TME is primed and the BBB is structurally compromised, the deployment of advanced, multi-target CAR T cells (engineered to resist PD-L1 exhaustion) and innate NK cell engagers (supercharged by IL-15 agonists) can be effectively utilized to execute durable, systemic immunological surveillance.

Finally, the clinical testing infrastructure is aligning with this biological reality. The brutal efficiency of Bayesian adaptive platforms like GBM-AGILE ensures that futile mechanisms and broad-spectrum failures are discarded rapidly, funneling resources strictly toward precision therapies that leverage specific molecular vulnerabilities. Furthermore, AI-driven radiomics and refined diagnostic criteria like RANO 2.0 ensure that effective inflammatory responses are no longer misclassified as disease progression. While a universal, single-agent cure for glioblastoma remains scientifically implausible, the highly choreographed, patient-specific combination of macroscopic surgical resection, biophysical barrier disruption, spatial-epigenetic inhibition, and synthetic immune activation presents a mathematically and biologically viable roadmap to transforming glioblastoma from a terminal diagnosis into a manageable, long-term remission.

Declarations

Funding: This research review received no external funding.

Conflicts of Interest: The author declares no conflicts of interest. Except my father diagnosed with the disease.

Data Availability Statement: Data sharing is not applicable to this article as no novel datasets were generated or analyzed during the preparation of this manuscript.

Works cited

1. Glioblastoma: Clinical Presentation, Multidisciplinary Management, and Long-Term Outcomes - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11719842/>
2. Glioblastoma—A Contemporary Overview of Epidemiology, Classification,

- Pathogenesis, Diagnosis, and Treatment: A Review Article - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12733314/>
3. CAR-T cell therapies are coming after glioblastoma: An overview of early phase clinical trials and future perspectives - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12886550/>
 4. Early Clinical Trial Results Show Promise for Glioblastoma Immunotherapy,
<https://hms.harvard.edu/news/early-clinical-trial-results-show-promise-glioblastoma-immunotherapy>
 5. The challenges and clinical landscape of glioblastoma immunotherapy - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11524205/>
 6. ASCO 2025: Dual-target CAR T-cell therapy slows growth of aggressive brain cancer,
<https://ecancer.org/en/news/26551-asco-2025-dual-target-car-t-cell-therapy-slows-growth-of-aggressive-brain-cancer>
 7. Drug resistance in glioblastoma: Challenges, mechanisms and therapeutic strategies (Review) - Spandidos Publications,
<https://www.spandidos-publications.com/10.3892/mco.2025.2913?text=fulltext>
 8. Emerging insights into the immunosuppressive tumor microenvironment and its implications for glioblastoma immunotherapy - Frontiers,
<https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2025.1665742/full>
 9. Current Standards of Care in Glioblastoma Therapy - NCBI,
<https://www.ncbi.nlm.nih.gov/books/NBK469987/>
 10. Glioblastoma Stem Cells at the Nexus of Tumor Heterogeneity, Immune Evasion, and Therapeutic Resistance - MDPI, <https://www.mdpi.com/2073-4409/14/8/562>
 11. Breakthrough in treatment approach showing promise in the fight against glioblastoma,
<https://cancerblog.mayoclinic.org/2025/01/16/breakthrough-in-treatment-approach-showing-promise-in-the-fight-against-glioblastoma/>
 12. Challenges and Opportunities in High-Grade Glioma Management and Imaging-Based Response Monitoring During Novel Immunotherapies - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12524075/>
 13. Updates in Glioma and Meningioma Classification – WHO 2021 & New Entities in the Literature - College of American Pathologists,
<https://www.cap.org/member-resources/articles/updates-in-glioma-and-meningioma-classification-who-2021-new-entities-in-the-literature>
 14. The 2021 World Health Organization Central Nervous System Tumor Classification: The Spectrum of Diffuse Gliomas - MDPI,
<https://www.mdpi.com/2227-9059/12/6/1349>
 15. The 2021 WHO Classification of Tumors of the Central Nervous System: a summary - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8328013/>

16. Definition of Glioblastoma Correlation and the New WHO 2021 GBM - American Journal of Neuroradiology,
<https://www.ajnr.org/content/ajnr/early/2024/07/11/ajnr.A8225.full.pdf>
17. The WHO 2021 Classification of Central Nervous System tumours: a practical update on what neurosurgeons need to know—a minireview - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9427889/>
18. IDH wild-type glioblastoma: Predictive value of standard-of-care (SOC) MRI for establishing MGMT promoter methylation status - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12321826/>
19. Niche-specific epigenetic interventions in the spatially ... - Frontiers,
<https://www.frontiersin.org/journals/molecular-biosciences/articles/10.3389/fmolb.2025.1704311/full>
20. The epigenetic mechanisms involved in the treatment resistance of glioblastoma - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11977385/>
21. Longitudinal multimodal profiling of IDH-wildtype glioblastoma ...,
<https://academic.oup.com/neuro-oncology/article/27/1/89/7889077>
22. Going viral: targeting glioblastoma using oncolytic viruses - Oxford Academic,
<https://academic.oup.com/immunotherapyadv/article/5/1/ltaf024/8212074>
23. Unraveling Glioblastoma Heterogeneity: Advancing Immunological ...,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12384581/>
24. Unraveling the immunosuppressive microenvironment of glioblastoma and advancements in treatment - Frontiers,
<https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2025.1590781/full>
25. Virus-Based Therapy Boosts Anti-Cancer Immune Responses to ...,
<https://www.dana-farber.org/newsroom/news-releases/2026/virus-based-therapy-boosts-anti-cancer-immune-responses-to-brain-cancer>
26. The Immunosuppressive Microenvironment of Glioblastoma: Mechanisms, Clinical Challenges and Future Directions - PubMed,
https://pubmed.ncbi.nlm.nih.gov/?term=XSTS2025183%2FAcademic+Enhancement+Support+Program+of+Hainan+Medical+University%5BGrants+and+Funding%5D&sort=date&sort_order=desc
27. Unraveling the immunosuppressive microenvironment of glioblastoma and advancements in treatment - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12119497/>
28. Within and beyond the tumor: Mechanisms of glioblastoma-induced immunosuppression | Neuro-Oncology Advances | Oxford Academic,
https://academic.oup.com/noa/article/7/Supplement_4/iv4/8249869
29. Highlights in IO: next-generation CAR-T therapy for glioblastoma,
<https://jitc.bmj.com/content/14/4/e014670>
30. Preliminary Clinical Trial Results Show 'Dramatic and Rapid' Regression of

- Glioblastoma after Next Generation CAR-T Therapy,
<https://www.massgeneral.org/news/press-release/clinical-trial-results-show-dramatic-regression-of-glioblastoma-after-next-generation-car-t-therapy>
31. Translational advancements in tumor vaccine therapies for glioblastomas - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12418593/>
 32. SurVaxM Shows Continued Promise in Glioblastoma as Phase IIb SURVIVE Trial Progresses,
<https://www.appliedclinicaltrials.com/view/survaxm-promise-glioblastoma-survive-trial>
 33. Roswell Park Announces Positive Progress in Clinical Trial of Novel Immunotherapy SurVaxM,
<https://physicianresources.roswellpark.org/news/roswell-park-announces-positive-progress-in-clinical-trial-of-novel-immunotherapy-survaxm>
 34. Study Details | NCT05163080 | SurVaxM Plus Adjuvant Temozolomide for Newly Diagnosed Glioblastoma (SURVIVE) | ClinicalTrials.gov,
<https://clinicaltrials.gov/study/NCT05163080>
 35. VBI Vaccines Announces New Tumor Response Data from Ongoing Randomized Controlled Phase 2b Study of VBI-1901 in Recurrent Glioblastoma Patients,
<https://www.vbivaccines.com/press-releases/vbi-vaccines-announces-new-tumor-response-data-from-ongoing-randomized-controlled-phase-2b-study-of-vbi-1901-in-recurrent-glioblastoma-patients/>
 36. Initial Data Shows 100% Disease Control in 5 Out of 5 Patients With Recurrent Glioblastoma With Two Patients in Near Complete Response Treated With ImmunityBio's ANKTIVA®, NK Cell Therapy Plus Optune Gio® Device,
<https://immunitybio.com/initial-data-shows-100-disease-control-in-5-out-of-5-patients-with-recurrent-glioblastoma-with-two-patients-in-near-complete-response-treated-with-immunitybios-anktiva-nk-cell-therapy-p/>
 37. Recent advances in Tumor Treating Fields (TTFields) therapy for glioblastoma | The Oncologist | Oxford Academic,
<https://academic.oup.com/oncolo/article/30/2/oyae227/7821273>
 38. NCT06556563 | EF-41/KEYNOTE D58: Phase 3 Study of Optune Concomitant With Temozolomide Plus Pembrolizumab in Newly Diagnosed Glioblastoma | ClinicalTrials.gov, <https://clinicaltrials.gov/study/NCT06556563>
 39. TIGER Study Reports New TTFields Therapy Survival Results for ...,
<https://www.novocure.com/tiger-study-reports-new-ttfields-therapy-survival-results-newly-diagnosed-glioblastoma-patients>
 40. Tumor Treating Fields: An Innovative Therapy for Glioblastoma and Other Solid Tumors - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12487835/>
 41. Focused Ultrasound for Glioblastoma (GBM): Clinical Results and Follow-On Trial in Taiwan,
<https://www.fusfoundation.org/posts/focused-ultrasound-for-glioblastoma-gbm-clin>

- [ical-results-and-follow-on-trial-in-taiwan/](#)
42. Study Details | NCT05864534 | Phase 2a Immune Modulation With Ultrasound for Newly Diagnosed Glioblastoma | ClinicalTrials.gov,
<https://clinicaltrials.gov/study/NCT05864534>
 43. Glioblastoma Clinical Trial: Focused Ultrasound Blood-Brain Barrier Opening is Safe, Provides Possible Survival Benefit,
<https://www.fusfoundation.org/posts/glioblastoma-clinical-trial-focused-ultrasound-blood-brain-barrier-opening-is-safe-provides-possible-survival-benefit/>
 44. Disrupting the Blood-Brain Barrier to Improve Glioblastoma Treatment | Spotlight | Advances in Neurology and Oncology - NewYork-Presbyterian Hospital,
<https://www.nyp.org/advances/article/disrupting-the-blood-brain-barrier-to-improve-glioblastoma-treatment>
 45. Glioblastoma - Focused Ultrasound Foundation,
<https://www.fusfoundation.org/diseases-and-conditions/glioblastoma/>
 46. The Top Brain Cancer News of 2025 | Targeted Oncology,
<https://www.targetedonc.com/view/the-top-brain-cancer-news-of-2025>
 47. Nanotherapy of Glioblastoma—Where Hope Grows - MDPI,
<https://www.mdpi.com/1422-0067/26/5/1814>
 48. Study Details | NCT03970447 | A Trial to Evaluate Multiple Regimens in Newly Diagnosed and Recurrent Glioblastoma | ClinicalTrials.gov,
<https://clinicaltrials.gov/study/NCT03970447>
 49. GBM AGILE - National Brain Tumor Society,
<https://braintumor.org/research/clinical-trials/gbm-agile/>
 50. Evaluation of Regorafenib in Newly Diagnosed and Recurrent ...,
<https://ascopubs.org/doi/10.1200/JCO-25-01137>
 51. Evaluation of VAL-083 in GBM AGILE, a phase 3 registration platform trial for newly diagnosed and recurrent glioblastoma. - ASCO,
<https://www.asco.org/abstracts-presentations/231460>
 52. GBM Agile: A Revolutionary New Adaptive Trial Platform - NFCR,
<https://www.nfcr.org/gbm-agile/>
 53. Navigating Neuro-Oncology: Pseudoprogression, Glioma & RANO 2.0 - merit cro,
<https://meritcro.com/navigating-neuro-oncology-pseudoprogression-glioma-and-rano-2-0/>
 54. VAL-083 Fails to Outperform SOC in Glioblastoma - OncLive,
<https://www.onclive.com/view/val-083-fails-to-outperform-soc-in-glioblastoma>
 55. RANO 2.0: critical updates and practical considerations for radiological assessment in neuro-oncology - BRAINLIFE.ORG,
https://www.brainlife.org/fulltext/2025/Sakata_A250630_JpnJRadiol.pdf
 56. RANO 2.0 criteria: concepts applicable to the neuroradiologist's clinical practice - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11493521/>
 57. RANO 2.0: critical updates and practical considerations for radiological assessment

in neuro-oncology - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12479667/>
58. IMG-143. AI-Guided RANO 2.0 Assessment and Pseudoprogression Incidence in
Glioblastoma: A Comparative Analysis of Bidimensional and Volumetric
Measurements | Neuro-Oncology | Oxford Academic,
https://academic.oup.com/neuro-oncology/article/27/Supplement_5/v309/8319312